

## INTERPOLATION AND SMOOTHING OF MAPPED CIRCULAR DATA

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unit vector.

**Summary:** We present a method of interpolation and smoothing based on geostatistical methods which is appropriate for unit vector data available at a number of spatial locations. The method takes account of irregular spacing in the data locations, and also of measurement error in the data. As an example of the use of the technique, we estimate smoothed values of one of the phase parameters of a daily rainfall model, and compare the accuracy of the method with that of a method based on inverse distance weighting.

### 1. INTRODUCTION

Circular data, that is, data taking values on the unit circle as opposed to the real line, arise in many fields. The most common examples are measurements of directions in two-dimensional space, such as in studies of animal navigation, wind direction or direction of magnetisation of rock specimens, or data arising from periodic phenomena, such as the hourly distribution of births in a hospital, or the time of year of maximum rainfall. Numerous examples are given in the texts by Mardia (1972), Batschelet (1981) and Upton and Fingleton (1989). Statistical techniques for circular data tend to be more computationally complex than their counterparts for data taking values on the real line. Notions of correlation, regression and bias are still the subject of discussion (Mardia (1975), Jupp and Mardia (1989)). In particular, the development of smoothing techniques for spatially distributed circular data is very much in its infancy. Such spatially distributed data

arise in a number of contexts in the earth sciences, for example, the direction of wind or of ocean currents measured at a number of locations, or the direction of trends of geological structures.

We review methods of interpolation and smoothing and suggest a new approach, based on geostatistical concepts.

This work was motivated by the need to interpolate the phase parameters of a daily rainfall model which was fitted to a number of sites in southern Africa (Zucchini & Adamson, 1984), and part of this data set is used to illustrate the technique.

## 2. THE DAILY RAINFALL MODEL

Zucchini and Adamson (1984) modelled daily rainfall at a number of sites in southern Africa using a model with two basic components; the occurrence of rain was modelled using a seasonal first-order Markov chain, and the amount of rain on wet days was modelled via a two-parameter distribution with a seasonally varying mean and a constant coefficient of variation. Thus, for each day of the year, three values are estimated; the probability of a wet day, given a preceding wet day, the probability of a wet day, given a preceding dry day, and the mean rainfall depth on wet days. Each site is thus represented by the following  $3 \times 365$  parameters:

$$Pr_{w/w}(t), \quad Pr_{w/d}(t), \quad \mu(t) \quad t = 1, 2, \dots, 365$$

plus the coefficient of variation.

These parameter values can be estimated, using maximum likelihood, at any site at which a sufficiently long rainfall record is available. For example,  $Pr_{w/d}(2)$  is estimated as the number of years in which day 1 was dry and day 2 was wet divided by the total number of years in which day 1 was dry. In practice we would expect the seasonal change in the parameters to be a smooth function of  $t$ , and Zucchini and Adamson found that a Fourier series approximation using only two harmonics provided an adequate model for

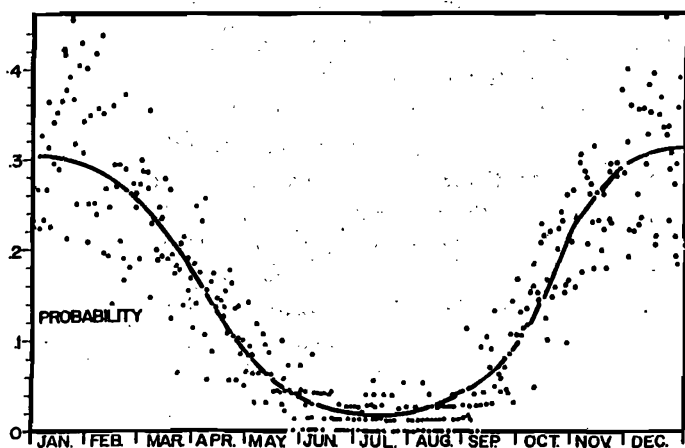


Figure 1. Estimated probability of a wet day following a dry day at Pretoria, based on a five-parameter model (solid line), compared with actual frequencies over a 76 year period.

sites throughout southern Africa. Thus for example,

$$(1) \quad \mu(t) = \alpha_0 + \sum_{i=1}^2 \alpha_i \cos(2\pi i(t - 1 - \beta_i)/365) \quad t = 1, 2, \dots, 365$$

so that the 365 values of  $\mu$  are replaced by the three amplitude parameters  $\alpha_0, \alpha_1, \alpha_2$  and two phase parameters  $\beta_1, \beta_2$ . For the two conditional probabilities,  $Pr_{w/w}(t)$  and  $Pr_{w/d}(t)$ , a similar Fourier series representation can be used, except that the Fourier series is fitted to the logits of the probabilities rather than to the probabilities themselves, in order to overcome the problem of negative probability estimates. Figure 1 shows the Fourier series fitted to the probability of a wet day following a dry day at Pretoria, together with the actual frequencies, based on a 76 year rainfall record. By using the Fourier series representation the total number of parameters at each site is reduced from  $(3 \times 365 + 1)$  to  $(3 \times 5 + 1)$ , and these 16 parameters were fitted by Zucchini and Adamson to every site in South Africa

with at least 20 years of daily rainfall records. A brief description of the model and the parameter estimation, together with some applications of the model, can be found in Zucchini *et al.* (1992).

In 1992, using more recent data, the original estimates were updated and a number of new sites were included, giving 3356 sites in all. Thus the available data set consists of the 16 parameter estimates at each of the 3356 sites.

In some parts of southern Africa there are very few locations at which daily rainfall records are available, and it is thus desirable to be able to interpolate the estimated model parameters from the existing sites to any given location, or to a grid of locations covering southern Africa. While a natural approach would be to consider the simultaneous estimation of all 16 parameters, using a multivariate technique, we here consider only the simpler problem of estimating individual parameters; some testing at individual sites suggests that such a univariate approach provides good estimates of the overall model.

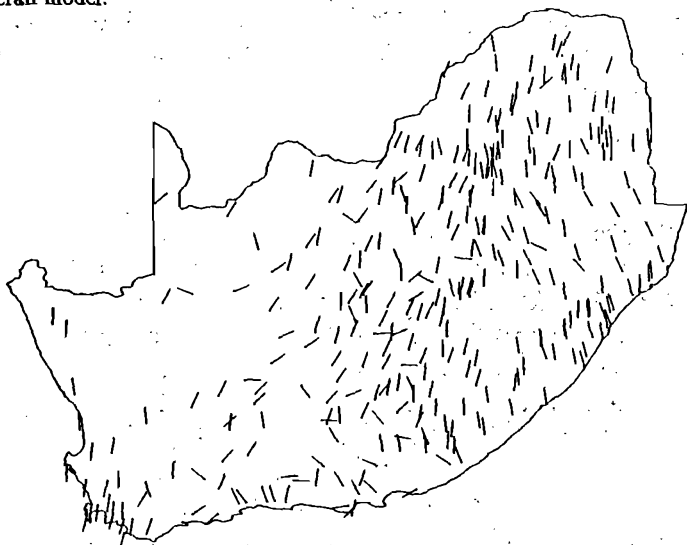


Figure 2. First phase parameter of  $Pr_{w/w^*}(t)$  at 325 randomly selected sites.

In this paper we focus on the specific task of estimating the phase parameters, such as  $\beta_1$  and  $\beta_2$  in equation (1), which are circular in nature. For example, the first phase parameter ( $\beta_1$  in equation (1)), which essentially represents the time of year at which the corresponding model component is a maximum, takes values between 0 and 365, while the second phase parameter can take values between 0 and 365/2. For example, for the Pretoria location shown in figure 1, the first phase parameter has a value of 9 (9th January), while the second phase parameter has a value of 106. Figure 2 shows a map of the first phase parameter for the logit of the probability of a wet day given a preceding wet day, at 325 sites, selected at random from the full set of 3356 sites. In this map, the small triangle marks the site location, while the direction of the line pointing away from the site location indicates the parameter value, with north representing the first of January. Thus the values in the northern part of the map reflect days mostly in December and January (summer rainfall area), while in the south-west corner of the map most of the values are in July (winter rainfall area). It can be seen that the distribution of the data locations is not even across the area; there is very little data in the north west or in the Lesotho highlands to the east, while in some areas, notably around major urban centres, there are fairly dense clusters of data. The map also indicates that there is a fair degree of spatial continuity in the data, so that some sort of spatial interpolation is feasible.

In estimating the parameter values at new locations we also need to take into account that the fitted model parameter values at the 3356 sites are themselves only estimates of the corresponding true parameter values, and are thus subject to error due to the finite length of the original daily rainfall record. It has not proved possible to obtain an estimate of this error by direct (analytic) means, and thus a parametric bootstrap procedure was used. That is, at each site, 100 samples of  $n$  years of data, where  $n$  was the length in years of the rainfall record at the given site, were simulated using the original parameter estimates, and the model parameters were re-estimated from each of the generated data sets; the variability of these 100

estimates provides a measure of the model parameter estimation error at that site. This error is thus not constant across the sites; in particular, it tends to be higher for sites at which the number of years of available rainfall record,  $n$ , is small, and also higher in areas where the rainfall is less frequent or more variable from year to year.

### 3. MEANS AND VARIANCES FOR CIRCULAR DATA

In this section we review some basic notation and summary statistics for circular data.

Circular data can be represented by points on a circle of unit radius. Where the data are not directions, as in the present application, the range of values can easily be mapped onto the circle; for example, in the case of the first phase parameters of the rainfall model, which take values between 0 and 365, we can multiply the values by  $2\pi/365$ . The mean of the data is then defined to be the direction of the resultant vector. That is, if we represent the data points by the unit vectors  $e_1, e_2, \dots, e_n$ , with polar co-ordinates  $(1, \theta_1), (1, \theta_2), \dots$  then the mean vector of the sample is given by

$$\mathbf{m} = \sum_{i=1}^n \mathbf{e}_i / n.$$

If we assign a unit mass to each data point in figure 3, then  $\mathbf{m}$  represents the centre of gravity of the data. The Cartesian co-ordinates of  $\mathbf{m}$  are  $\bar{x} = \sum \cos \theta_i / n$  and  $\bar{y} = \sum \sin \theta_i / n$  and the polar co-ordinates are  $(\bar{R}, \bar{\theta})$  where

$$\bar{R}^2 = \bar{x}^2 + \bar{y}^2$$

and

$$\tan \bar{\theta} = \sum \sin \theta_i / \sum \cos \theta_i$$

This has a singularity if  $\sum \sin \theta_i = \sum \cos \theta_i = 0$ , so that the centre of gravity is at the origin, and thus the resultant direction is not uniquely defined.

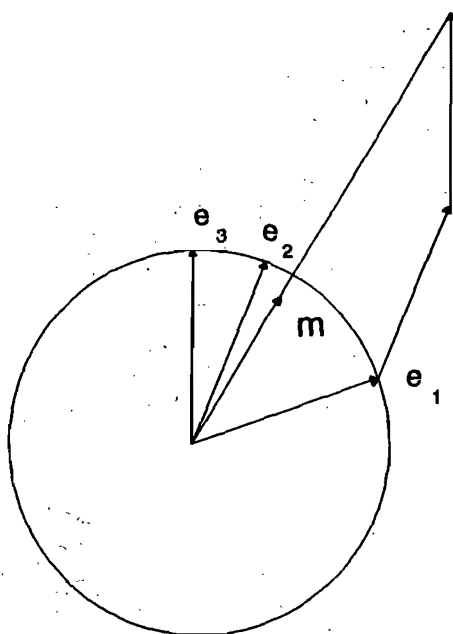


Figure 3. Calculation of mean of unit vectors.

It can be shown that

$$\bar{R} = \sum_{i=1}^n \cos(\theta_i - \bar{\theta})/n$$

and thus the measure  $[1 - \bar{R}]$  provides a measure of the variance, with properties which are in many ways analogous to those of the variance for non-circular data (Mardia, 1972).

To obtain a weighted mean with weights  $w_1, \dots, w_n$  it is natural to define this as the direction corresponding to the point with co-ordinates  $\bar{x} = \sum w_i \cos \theta_i / n$  and  $\bar{y} = \sum w_i \sin \theta_i / n$  which is equivalent to assigning the weights as masses to the data points and finding the centre of gravity as

before. Note that multiplying all the weights by a non-zero constant,  $l$ , does not affect the direction of the weighted mean, but changes the length of the mean vector  $\bar{R}$  by a factor  $l$ .

#### 4. REVIEW OF SMOOTHING METHODS FOR CIRCULAR DATA

Almost all of the published work on interpolation and smoothing of circular or spherical data has been directed towards the problem of data indexed in one dimension. Many of these methods are not truly vectorial in nature. For example, Parker and Denham (1979), studying the problem of plotting the position of the magnetic north pole at various geological times, suggested splining the three individual co-ordinates of the unit vector independently, and then re-normalizing to get unit vectors; this method could be extended to data available at locations in two dimensions using thin plate splines, but the method has been criticised by a number of authors, including Watson (1985) and Jupp and Kent (1987), as it ignores the geometry of the sphere. Other methods, such as that of Clark and Thompson (1984) which involves projecting the data onto a tangent plane, do not work well unless the data are restricted to a portion of the sphere or circle, and produce fitted paths which are not invariant to change of origin. Since the data set shown in figure 2 clearly takes values throughout the full range of angles on the circle, such methods are not suitable for our application.

The method proposed by Jupp and Kent (1987) does not readily extend to spatial data as they require that the circular data be 'unwrapped' onto the real line; it is not clear that this can be done consistently for data indexed in two dimensions.

Another feature of almost all the methods suggested for smoothing circular or spherical data is that they are spline based, and Watson (1985) suggests that, even in the relatively simple case when the data locations are on a line, rather than a plane, there is a need to penalise excessive spiralling of the estimated values around the unit circle, in addition to the usual roughness penalty used for splines, and this leads to a considerably more complex problem.

Watson (1985) is perhaps the first to discuss the problem of interpolating and smoothing circular or spherical data indexed in two dimensions, that is, data available at a number of spatial locations, and he outlines a number of possible approaches to these problems. For the interpolation problem he suggests the use of a weighted average of the data points, with the weights chosen proportional to the inverse of the square of the distance from the point to be estimated. He goes on to outline a possible method for smoothing spatial directional data, based on calculating estimated values  $f(\mathbf{z}_1)$  to  $f(\mathbf{z}_n)$  at the given spatial locations  $\mathbf{z}_i$  ( $i = 1, 2, \dots, n$ ), with corresponding observed values  $\theta_i$ , to maximize

$$\sum_{i=1}^n k_i \cos(\theta_i - f(\mathbf{z}_i)) + \tau \sum_{1 \leq i < j \leq n} \cos(f(\mathbf{z}_i) - f(\mathbf{z}_j))/d_{ij}$$

where  $k_i$  is some inverse function of the measurement error variance of the  $i$ -th data point,  $d_{ij}$  is a some monotonic increasing function of distance between the  $i$ -th and  $j$ -th data points and  $\tau$  is a smoothing parameter. Watson does not give full details of the method; in particular, the discussion does not explain how the method generates estimates at points other than the original data locations  $\mathbf{z}_i$ ; typically estimates are required at a grid of points. Also, if the original data locations are clustered in space, then excessive weight will be given to 'high density' areas. This criticism applies also to the simple inverse distance method mentioned above.

Mendoza (1986) has subsequently implemented a method of smoothing circular data available at locations on a plane and illustrates its use to smooth data on the cross-bedding directions of sandstone. His method, which is rather similar to Watson's except that it uses a spline-based measure of smoothness, finds  $f(\mathbf{z})$  to minimize

$$\sum_{i=1}^n w_i (1 - \cos(\theta_i - f(\mathbf{z}_i))) + \lambda R_2(f)$$

where  $\theta_i$  is the observed angle and  $f(\mathbf{z}_i)$  the smoothed angular value at the  $i$ -th location,  $w_i$  is a weighting factor for the  $i$ -th data point and  $R_2(f)$  is

a measure of roughness of the function  $f$ . The roughness criterion is similar to that used in spline-smoothing and is given by

$$R_2(f) = \iint \left[ \left( \frac{\partial^2 f}{\partial x^2} \right)^2 + 2 \left( \frac{\partial^2 f}{\partial x \partial y} \right)^2 + \left( \frac{\partial^2 f}{\partial y^2} \right)^2 \right] dx dy.$$

The calculation of  $R_2(f)$  requires that the data first be translated from unit vectors to angles, so that it is first necessary to choose the value  $\theta_i$  to represent each data point. Clearly these angles are only determined up to multiples of  $2\pi$ , and Mendoza suggests that they should be chosen 'so that observations are not rougher than they should be'. Thus, for example, a sequence of adjacent values (in degrees) of 240 300 350 30 is re-expressed as 240 300 350 390. This may not be easy to achieve consistently for spatial data. For example, figure 4 shows some data, taken from the south-west corner of the larger map of actual data in figure 6, in which smoothing the data along the path indicated by the dotted line from the point A (192 degrees) to the point B leads to a labelling of 381 degrees for point B whereas smoothing the data along the path indicated by the dashed line leads to a

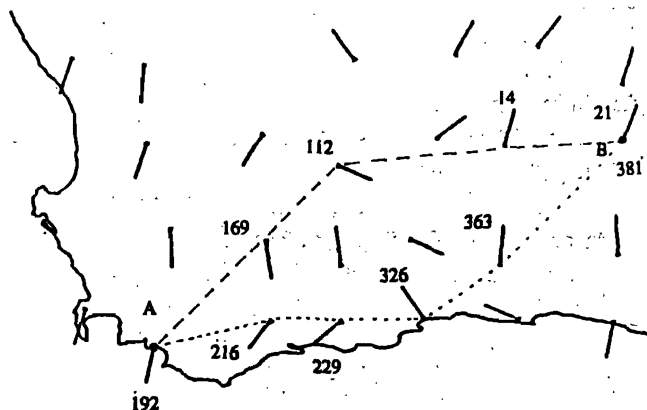


Figure 4. Unwrapping of circular data values along two different paths.

value of 21 degrees for the same point; the lack of a natural ordering of points in two-dimensional space means that appropriate values may not be uniquely defined.

Ideally what is required for applications such as the one discussed in this paper, is a method of smoothing circular data (with interpolation as a limiting case), which is truly vectorial in nature and which is appropriate for data indexed in two dimensions, with possible clustering of the data locations in some areas. Further, we require a method which can make allowance for the fact that the variance of the measurement error in the data is not constant across all data points.

## 5. A GEOSTATISTICAL APPROACH FOR CIRCULAR DATA

The geostatistical technique known as kriging is a popular method for interpolating non-circular spatial data and has a number of advantages over other simpler methods, such as inverse-distance (kernel) smoothing, particularly if the data are irregularly spaced. A number of empirical studies (Creutin & Obled (1982), Tabios & Salas (1985)) have shown kriging to compare favourably with other methods, and it has also been shown that there is a close relationship between kriging and spline-smoothing (Cressie, 1991).

In kriging, the data is modelled as a realization of a stochastic process with constant variance, with the covariance generally assumed to be a function of distance only and estimated from the empirical covariance function or semi-variogram. The mean value of the process may be assumed to be stationary or may be modelled as a function of the spatial locations as in so-called *universal* kriging. It is well known (Cressie, 1991) that for this model the minimum variance unbiased linear predictor of the value at a location  $z_0$ , based on values  $v_1$  to  $v_n$  at locations  $z_1$  to  $z_n$ , is of the form

$$(2) \quad \hat{v}_0 - \hat{\mu}_0 = s' K^{-1} (v - \hat{\mu})$$

where  $s_i = \text{cov}(v_i, v_0)$  and  $K$  has elements  $k_{ij} = \text{cov}(v_i, v_j)$ .

Without the matrix  $K$  in equation (2) this would be essentially an inverse-distance or kernel-smooth solution with the weights  $s_i$  based on the covariances which in turn are modelled as a function of distance; the appearance in the solution of the matrix  $K$  of covariances ensures that the relative spatial locations of the data points are taken into account, and that clusters of closely-spaced, and thus highly correlated, data points are not given excessive weight in the estimation.

When measurement error is present in the data a prediction of the (unobservable) error-free values is required and thus it is appropriate to smooth rather than to interpolate the original data. Cressie (1988) shows that in this case the vector  $s$  in the equation above must be replaced by the vector of covariances between the data values and the error-free value at location  $z_0$ ; this necessitates an estimate of the measurement error variance.

In view of the advantages and the generality of the kriging technique for non-circular data it is natural to attempt to extend the kriging method to the problem of interpolation and smoothing of circular data.

In the sections which follow, we first define a method of measuring the spatial covariance of circular data which is analogous to the usual semi-variogram. We then formulate a linear predictor for circular data based on minimizing the expected mean square error, as in kriging. This leads to the necessity of maximizing the expectation of a non-linear function of random variables, for which an exact solution is not available without making further distributional assumptions. We consider an approximate solution and compare it with the alternative option of simple inverse distance weighting for the estimation of one of the phase parameters of the daily rainfall model described in section 2.

## 6. MEASURING THE SPATIAL COVARIANCE FOR CIRCULAR DATA

A number of measures of association have been proposed for circular data, and reviews can be found in Jupp and Mardia (1989) and Breckling (1989).

For our purposes here it suffices to find a measure of the relationship between two circular variables with the same distribution, and, in particular, with the same mean, and for this purpose we define

$$(3) \quad \gamma(h) = E_{(d_{ij}=h)} [1/2(1 - \cos(\theta_i - \theta_j))]$$

as a circular analogue of the usual semi-variogram, where the expectation is over all locations  $i$  and  $j$  with separation distance  $h$ . We note that this is effectively 1 minus the covariance measure proposed by Breckling; the rotation matrix in Breckling's measure being omitted here since  $\theta_i$  and  $\theta_j$  are assumed to have the same mean. This assumption is analogous to that made in the use of the semi-variogram in ordinary kriging. The circular semi-variogram as thus defined takes values between 0 and 1.

In order to study the empirical circular semi-variogram as a function of separation distance we can plot the average of the values  $(1 - \cos(\theta_i - \theta_j))/2$  for all pairs of points with a given separation distance as a function of the separation distance. When the data is measured with error, this empirical semi-variogram will be increased by an amount depending on the error variance. We show in the appendix that if we observe  $\tau_i$  where  $\tau_i = \theta_i + \epsilon_i$  and if the error terms  $\epsilon_i$  are independent and have a symmetric distribution with mean zero, then

$$E[\cos(\tau_i - \tau_j)] = E[\cos(\theta_i - \theta_j)] E[\cos \epsilon_i] E[\cos \epsilon_j].$$

If an estimate of the error variances  $1 - E[\cos \epsilon_i]$  is available, then the empirical semi-variogram for the observed  $\tau$  values may be adjusted by dividing through by the appropriate values of  $E[\cos \epsilon_i]$  and  $E[\cos \epsilon_j]$  for each pair of data points to get an estimate of the semi-variogram of the underlying variable  $\theta$ . For the daily rainfall model an error estimate is available for each site by the use of the bootstrap as described in section 2; figure 5 shows the unadjusted and adjusted semi-variogram for the first phase parameter of the logit of the probability of a wet day given a preceding wet day. The graphs both increase with distance initially and then level off, as would be expected. The distance at which the graph levels off is indica-

tive of the range of spatial correlation of these values. The negative values in the lower graph result from the adjustment for measurement error; clearly, in fitting a model to such data one would require all the fitted values to be positive.

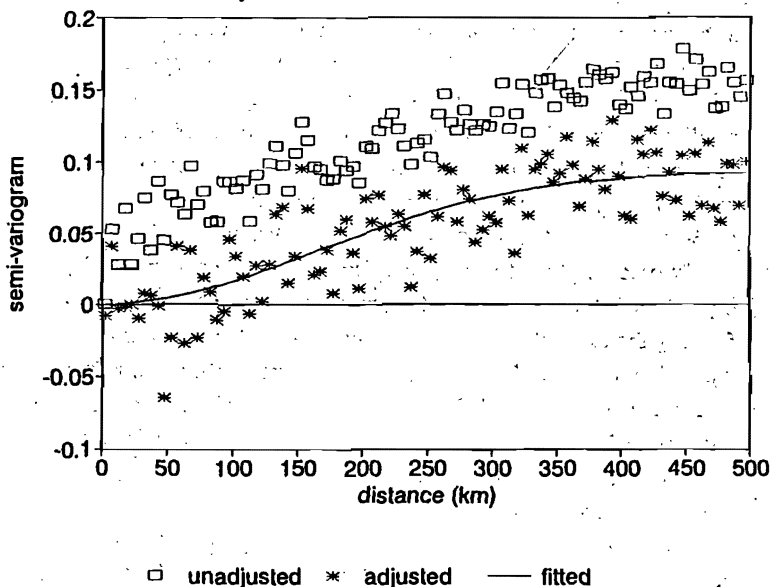


Figure 5. Circular semi-variogram for first phase parameter of  $Pr_{w/w}(t)$

If we assume, as is usual in geostatistics, that the covariance between two points is a function of distance only (or possibly of distance and direction), then we can model the spatial covariance as a function of distance by fitting a function to the empirical semi-variogram. Common geostatistical practice is to fit one of a number of standard functions, such as the

'spherical' or 'exponential' semi-variogram functions, which ensure a positive-definite covariance. The model is fitted to the empirical semi-variogram using one of a number of possible methods as described in the review by Zimmerman and Zimmerman (1991); alternatively some users prefer to fit the parameters of the semi-variogram model by cross-validation, that is, by selecting those parameters which minimize the least squares error in the kriging estimation of each of the data points from the others.

## 7. KRIGING FOR CIRCULAR DATA

We consider a model equivalent to that used in *ordinary* kriging, that is, we assume the data are a realization of a stochastic process with common mean and variance, and that the covariance, measured by  $E[\cos(\theta_i - \theta_j)]$ , is a function of distance only.

Given unit vectors  $e_1, e_2, \dots, e_n$ , at spatial locations  $x_1, x_2, \dots, x_n$ , we seek an estimator, based on the weighted sum  $\sum w_i e_i$ , of the value at some location  $x_0$ . Specifically, if  $\sum w_i e_i$  is written in polar form as  $(R_0, \hat{\theta}_0)$  then we seek  $w$  such that  $E[1 - \cos(\hat{\theta}_0 - \theta_0)]$  is a minimum, where  $\theta_0$  is the unknown angular value at location  $x_0$ . The use of the function  $1 - \cos(\theta - \hat{\theta})$  as a measure of distance is common in circular statistics, and is analogous to the usual least squares criterion (Fisher & Lewis, 1985).

Write  $e_i = (x_i, y_i)'$  and let  $\theta_i = \arccos(x_i) = \arcsin(y_i)$  and  $e_0 = (x_0, y_0)'$  with  $\theta_0 = \arccos(x_0) = \arcsin(y_0)$ .

Then we have that

$$\sin \hat{\theta}_0 = \sum_{i=1}^n w_i y_i / R_0 = \sum_{i=1}^n w_i \sin \theta_i / R_0$$

and

$$\cos \hat{\theta}_0 = \sum_{i=1}^n w_i x_i / R_0 = \sum_{i=1}^n w_i \cos \theta_i / R_0$$

where  $R_0$  is the length of the vector  $\sum w_i e_i$ , so that

$$\begin{aligned}
R_0^2 &= \left[ \sum_{i=1}^n w_i x_i \right]^2 + \left[ \sum_{i=1}^n w_i y_i \right]^2 \\
&= \sum_{i=1}^n \sum_{j=1}^n w_i w_j \cos \theta_i \cos \theta_j + \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sin \theta_i \sin \theta_j \\
&= \sum_{i=1}^n \sum_{j=1}^n w_i w_j \cos (\theta_i - \theta_j) \\
&= \mathbf{w}' Q \mathbf{w}
\end{aligned}$$

where  $q_{ij} = \cos(\theta_i - \theta_j)$ .

Now

$$\begin{aligned}
\cos(\hat{\theta}_0 - \theta_0) &= \cos \hat{\theta}_0 \cos \theta_0 + \sin \hat{\theta}_0 \sin \theta_0 \\
&= 1/R_0 \left[ \sum_{i=1}^n w_i (\cos \theta_i \cos \theta_0 + \sin \theta_i \sin \theta_0) \right] \\
&= \mathbf{w}' \mathbf{c} / \sqrt{(\mathbf{w}' Q \mathbf{w})}
\end{aligned}$$

where  $c_i = \cos(\theta_i - \theta_0)$ .

Thus in order to minimize  $E[1 - \cos(\hat{\theta}_0 - \theta_0)]$  we need to find  $w_1, \dots, w_n$  to maximize

$$(4) \quad E[\mathbf{w}' \mathbf{c} / \sqrt{\mathbf{w}' Q \mathbf{w}}]$$

For the case when the data are measured with error, so that we observe  $\tau_i = \theta_i + \epsilon_i$  instead of  $\theta_i$ , a similar argument shows that

$$(5) \quad E[\cos(\theta_0 - \theta_0)] = E[\mathbf{w}' \mathbf{c}^* / \sqrt{\mathbf{w}' \mathbf{Q}^* \mathbf{w}}]$$

where  $c_i^* = \cos(\tau_i - \theta_0)$  and  $q_{ij}^* = \cos(\tau_i - \tau_j)$ .

Note that we have not introduced any constraint to ensure unbiasedness; we seek rather an estimator which minimizes the expected error, measured as a function of the cosine of the difference of the two angles.

It is clear that the solution will be unique only up to a constant multiplier; that is, if  $\mathbf{w}$  is a solution, then so is  $l\mathbf{w}$  for any non-zero constant  $l$ . In order to make the solution unique, we might introduce a constraint, such as  $\sum w_i = 1$ , or  $E[\mathbf{w}' \mathbf{Q} \mathbf{w}] = 1$ . Note that the constraint  $\mathbf{w}' \mathbf{Q} \mathbf{w} = 1$  is not an appropriate choice, since then the  $w_i$  will depend on the data values, so that, for example, we would not have  $E[\mathbf{w}' \mathbf{c}] = \sum w_i E[c_i]$ .

We may assume that the expected values of the elements of  $\mathbf{Q}$  and  $\mathbf{c}$  can be estimated from the modelling of the spatial covariance function; however, this is not of immediate help, since the expression to be maximized, as given by equation (4), cannot be expressed directly as a function of these, so that the usual method of maximization used in kriging does not apply directly in this case.

If we use a first-order Taylor series expansion of  $E[\mathbf{w}' \mathbf{c} / \sqrt{\mathbf{w}' \mathbf{Q} \mathbf{w}}]$ , so that we approximate it by  $\mathbf{w}' \mathbf{s} / \sqrt{\mathbf{w}' \mathbf{K} \mathbf{w}}$ , where  $k_{ij} = E[q_{ij}]$  and  $s_i = E[c_i]$ , and use the uniqueness constraint  $E[\mathbf{w}' \mathbf{Q} \mathbf{w}] = \mathbf{w}' \mathbf{K} \mathbf{w} = 1$  then we can use the Lagrange multiplier approach to find the optimal values of the  $w_i$ . With the chosen uniqueness constraint, the function to be maximized becomes simply  $\mathbf{w}' \mathbf{s}$ , so if we set

$$G = \mathbf{w}' \mathbf{s} - \lambda(\mathbf{w}' \mathbf{K} \mathbf{w} - 1)$$

then

$$\frac{\partial G}{\partial \mathbf{w}} = \mathbf{s} - 2\lambda \mathbf{K} \mathbf{w}$$

and

$$\frac{\partial G}{\partial \lambda} = \mathbf{w}' \mathbf{K} \mathbf{w} - 1$$

which leads to the solution

$$(6) \quad w = \frac{K^{-1} s}{\sqrt{s' K^{-1} s}} = \frac{K^{-1} s}{r}$$

where  $r = \sqrt{s' K^{-1} s}$  is a scalar normalizing constant.

For the case where the data include measurement error, we can replace  $K$  and  $s$  above with  $K^*$  and  $s^*$  where  $k_{ij}^* = E[q_{ij}^*]$  and  $s_i^* = E[c_i^*]$  with  $c_i^*$  and  $q_{ij}^*$  as defined in equation (5).

We note that if measurement error is not included then the estimator is an exact interpolator since, if the location  $x_0$  is the same as one of the data locations  $x_i$ , then the vector  $s$  will be identical to one of the rows/columns of the matrix  $K$ , leading to a weight of 1 for the corresponding data point and zero weights for all other points.

Since the solution given by (6) is based on a first-order approximation of the expression in (4) there is no guarantee that the solution will be close to the true optimum. A better approximation might be obtained by considering higher-order moments. Instead of pursuing this approach we consider the form of the solution given by (6) and note that it is, apart from the normalizing constant  $r$ , exactly analogous to the solution of the usual (non-circular) problem of *simple kriging*. The form of this solution is intuitively appealing, in that it gives more weight to those data values which are close (in space) to the point to be estimated (via  $s$ ) and gives less weight to points which are clustered with other data points (via  $K^{-1}$ ). Thus a solution of the form  $\bar{K}^{-1} \bar{s}$ , where the elements of  $\bar{K}$  and  $\bar{s}$  are a function of the inter-point distances, can be justified in its own right as an extension of simple kernel-smooth methods with advantages for clustered data. It is thus of interest to see how well such a method performs in practice.

In the next section we therefore investigate this approach using some of the data from the rainfall model described in section 2, and compare the results obtained by using the weights given by equation (6) with the simpler method of inverse distance weighting.

## 8. COMPARISON OF METHODS

For illustration we take as our data set the first phase parameter,  $\beta_1$ , of the Fourier series for the logit of the probability of a wet day on day  $t$ , given a wet day on day  $(t-1)$ . In order to have a separate test data set for evaluating the methods, we have first selected, at random, one data location from every fourth half-degree grid square, thus giving a set of 101 data points which lie approximately on a regular grid (figure 6). From the remaining data set of 3255 sites we have selected at random one in every ten to be used as the available data set for the estimation, giving the data set shown in figure 2. The method of selection of these 325 points ensures that their spatial coverage and clustering pattern are similar to those of the full data set, while the decision to use only part of the full data set for this exercise was determined partly by the need to be able to display all the points clearly on a map, but also for the reason that, if we use too dense a data set, almost any interpolation method will give excellent results.

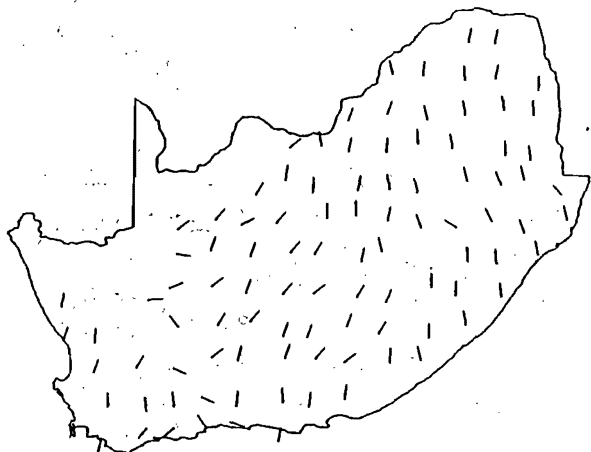


Figure 6. First phase parameter of  $Pr_{w/w_t}(t)$  at 101 selected test sites.

Based on the empirical semi-variogram adjusted for the measurement error in the data, as shown in figure 5, we have fitted a Gaussian model using the weighted least squares method of Cressie (1985), which gives most weight to the shorter lag distances and which also downweights lag distances which are based on a small number of data pairs. The Gaussian model is given by the function

$$a(1 - \exp(-d^2/r^2))$$

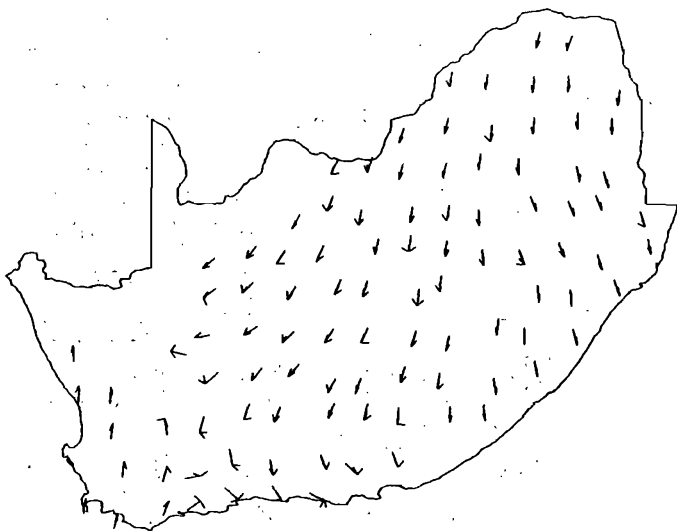
where  $d$  is the distance in kilometres. The resulting parameter estimates were 0.0929 for  $a$  and 233 for  $r$ .

Using this fitted function, each point in the test data set was estimated from the data values using the first-order approximation of the kriging solution as described in the previous section (equation (6)). Values were also estimated using the simple inverse squared distance weighting as described by Watson (1985). In both methods a search radius of 300 kilometres was used, that is, only points within 300 kilometres of the point to be estimated were included in calculation. Apart from reducing computation time, this will largely compensate for the fact that we have made no attempt to model or remove any trend in the data prior to using the kriging method.

## 9. DISCUSSION OF RESULTS

As discussed in section 7, the criterion which we wish to minimize is  $E[1 - \cos(\hat{\theta}_0 - \theta_0)]$ . The comparison described in the previous section resulted in an average of the error terms  $[1 - \cos(\hat{\theta}_i - \hat{\theta}_i)]$  (averaged over the 101 test data points) of 0.0988 for the kriging method and 0.1318 for the inverse distance method. There are several reasons why the kriging method may give better results than the inverse distance method. One is, of course, that in using the inverse distance method we have made no attempt to optimize the particular inverse distance function used; it would be possible to use a parametric function of distance, with the parameter value controlling the effective bandwidth selected, for example, by cross validation,

but this to some extent reduces the main advantage of the inverse distance method, namely its simplicity. Another possible reason for the superiority of the kriging method is that the inverse distance method does not take account of clustering in the data; however, a study of the data values in figure 2 suggests that this is probably not of great consequence for this particular data set, where the clustered data points generally have similar values to the more isolated points in the same vicinity. A third reason for the superiority of the kriging method is its explicit use of the error variance of the data; the inverse distance method will give relatively high weight to the few points which are closest to the point to be estimated even if those data points have high measurement error, whereas the kriging method will adjust for this; this is quite important in the present application where the error variance, as measured via the bootstrap procedure, was relatively high at some sites.



**Figure 7.** Estimated first phase parameter at 101 test sites compared with approximate 95% confidence interval for true parameter.

In comparing individual estimated values with the original values in the test data set one must bear in mind that even the values in the test data set are subject to the parameter estimation errors mentioned in section 2. Therefore, in plotting a map of the values estimated by the kriging method (figure 7) we have included for comparison, not the original data values, but a range of values (indicated by the two short lines emanating from each site in the figure) which represent a range of  $\theta_i \pm \varepsilon_i$ , where  $\theta_i$  is the original data value at that site and  $\varepsilon_i = \arccos(\sum_j \cos(\theta_{ij} - \theta_i))$ , where the  $\theta_{ij}$  are the individual bootstrap estimates used in estimating the measurement error, as described in section 2. For our data set this range corresponds roughly to an approximate 95% confidence interval for data having a von Mises distribution, based on the formulae given in section 9.6 of Upton and Fingleton (1989).

It can be seen in figure 7 that the fit of the estimated values is generally good, except for five sites which lie in the area of change-over between the winter rainfall area in the south-west and the summer rainfall area further to the north and east. The data set is relatively sparse in this area; clearly more data points are needed for accurate estimation in this region.

Table 1. Average error as a function of sill (a) and range (r) parameters.

| r   | a      |        |        |        |        |
|-----|--------|--------|--------|--------|--------|
|     | 0,07   | 0,08   | 0,09   | 0,10   | 0,11   |
| 100 | 0,1083 | 0,1081 | 0,1080 | 0,1078 | 0,1077 |
| 150 | 0,1069 | 0,1072 | 0,1075 | 0,1076 | 0,1077 |
| 200 | 0,1013 | 0,1019 | 0,1024 | 0,1029 | 0,1034 |
| 250 | 0,0964 | 0,0967 | 0,0971 | 0,0975 | 0,0978 |
| 300 | 0,0948 | 0,0947 | 0,0947 | 0,0947 | 0,0948 |
| 350 | 0,0961 | 0,0956 | 0,0953 | 0,0950 | 0,0948 |
| 400 | 0,0973 | 0,0969 | 0,0966 | 0,0963 | 0,0961 |
| 450 | 0,0973 | 0,0970 | 0,0968 | 0,0967 | 0,0965 |
| 500 | 0,0975 | 0,0970 | 0,0966 | 0,0963 | 0,0962 |

In the comparison described above the semi-variogram parameters were estimated directly from the empirical semi-variogram. However, since the solution given by (6) is only approximate there is no reason why these parameters should be optimal and it is likely that better results would be obtained if the parameters  $a$  and  $r$  were estimated via cross-validation. However, the cross-validation approach is more computationally intensive and it is thus of interest to test the sensitivity of the kriging method to the semi-variogram parameters. The estimation process was therefore repeated with a range of values of these parameters, and some of the results are shown in table 1, where the average error is given as a function of the 'sill' and 'range' parameters,  $a$  and  $r$  respectively. It is apparent from the results that the method is not particularly sensitive to the choice of parameters, as the error surface is fairly flat in the region of the parameter values used, although the global optimum (0,0942) was found at values of roughly 0,20 for  $a$  and 350 for  $r$ . In fact, a detailed search suggests that the error surface may well have a number of local minima, for example in the region of  $a = 0,09$  and  $r = 300$ , a fact that has been noted by other researchers in connection with maximum likelihood estimation of spatial covariance functions (Warnes & Ripley, 1987). Thus it would seem that, for this data set at least, using cross-validation to estimate optimal parameters is not likely to give much improvement over the computationally quicker method used here, based on modelling the empirical semi-variogram.

## 10. MEASURE OF ESTIMATION ERROR

One of the often-quoted advantages of the kriging method is that it provides an estimate of the error of estimation, although some writers have commented that the error estimate tends to be an under-estimate, based as it is on an estimated covariance model.

For the estimator given by equation (6) we have

$$\begin{aligned}
E[\cos(\hat{\theta}_0 - \theta_0)] &= E[\mathbf{w}'\mathbf{c}/\sqrt{\mathbf{w}'\mathbf{Q}\mathbf{w}}] \\
&= E\left[\frac{\mathbf{s}'\mathbf{K}^{-1}\mathbf{c}/\tau}{\sqrt{(\mathbf{s}'\mathbf{K}^{-1}/\tau)\mathbf{Q}(\mathbf{K}^{-1}\mathbf{s}/\tau)}}\right] \\
&= E\left[\frac{\mathbf{s}'\mathbf{K}^{-1}\mathbf{c}}{\sqrt{\mathbf{s}'\mathbf{K}^{-1}\mathbf{Q}\mathbf{K}^{-1}\mathbf{s}}}\right] \\
&\approx \sqrt{\mathbf{s}'\mathbf{K}^{-1}\mathbf{s}}
\end{aligned}$$

using again a first-order Taylor series approximation.

A similar expression, replacing  $\mathbf{s}$  with  $\mathbf{s}^*$  and  $\mathbf{K}$  with  $\mathbf{K}^*$  holds for the case where the data is measured with error.

In the example described in this paper we have, for simplicity, assumed that the covariance model is constant throughout the entire area, and thus the kriging estimation error at any point will reflect the configuration of data points in that locality and the inherent measurement error of those data values. It may be more realistic to assume that the spatial covariance is not constant throughout the entire area, and to fit separate covariance models in various areas, or even to use a moving window method (Haas, 1990), where the semi-variogram is re-estimated in the neighbourhood of each point to be estimated. This leads to a considerable increase in computational effort, but may give better estimates and more realistic estimates of error.

It is difficult with our data set to check the estimation errors, since we do not know the exact values of  $\theta_0$  at the test-points but only  $\tau_0 = \theta_0 + \epsilon_0$ . However, an examination of the estimation errors as given by  $\cos(\hat{\theta}_0 - \tau_0)$  for the 101 test points showed that none exceeded the approximate expectation as given by the formula above by more than a small amount, with the notable exception of five points in the change-over area between the summer and winter rainfall regions, where the observed error was much greater than predicted. This again suggests that it might be more realistic to use locally computed semi-variogram models.

## 11. CONCLUSION

We have presented a method of interpolation and smoothing for circular data based on the kriging approach. This method is independent of the choice of origin of the data and is fully vectorial in nature so that it does not rely on having a data set which is restricted to one part of the unit circle. The illustration given above suggests that the method has advantages over simpler methods such as inverse distance weighting in that it accounts for both clustering and measurement error in the data, and uses a distance measure based on the spatial continuity shown by the data values. The method also allows for an approximate measure of the estimation error. The computational effort is similar to that of simple kriging, and thus should not be more than that of spline-based methods.

Although the example given in this paper involved circular data, the method applies equally to unit vectors on the sphere.

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## APPENDIX

Suppose that we observe angular values  $\tau_i$  such that

$$\tau_i = \theta_i + \epsilon_i$$

where the angles  $\epsilon_i$  represent measurement error so that we may assume that the values of  $\epsilon_i$  at different sites are independent of one another and also of the values  $\theta_i$ . We also assume that the distribution of  $\epsilon_i$  is symmetric with mean zero so that  $E[\sin \epsilon_i] = 0$ . Then

$$\begin{aligned}
E[\cos(\tau_i - \tau_j)] &= E[\cos(\theta_i + \epsilon_i - \theta_j - \epsilon_j)] \\
&= E[\cos(\theta_i - \theta_j)\cos(\epsilon_i - \epsilon_j) - \sin(\theta_i - \theta_j)\sin(\epsilon_i - \epsilon_j)] \\
&= E[\cos(\theta_i - \theta_j)(\cos \epsilon_i \cos \epsilon_j + \sin \epsilon_i \sin \epsilon_j) \\
&\quad - \sin(\theta_i - \theta_j)(\sin \epsilon_i \cos \epsilon_j - \cos \epsilon_i \sin \epsilon_j)] \\
&= E[\cos(\theta_i - \theta_j)\cos \epsilon_i \cos \epsilon_j] \\
&= E[\cos(\theta_i - \theta_j)]E[\cos \epsilon_i]E[\cos \epsilon_j].
\end{aligned}$$

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